Deafness is for ever, not just for Christmas

Health Claims Forum

8th December 2011

Norman Haacke MA MB BChir FRCS FRCSI
Consultant ENT Surgeon

Formerly:
Consultant ENT Surgeon and Senior Lecturer
Southampton University Hospitals and Wessex Neurological Centre,
Medical Director, Wessex Regional Hearing and Balance Centre,
Director, South of England Cochlear Implant Centre

What is sound

A vibrating object (guitar string, loudspeaker cone, tuning fork)

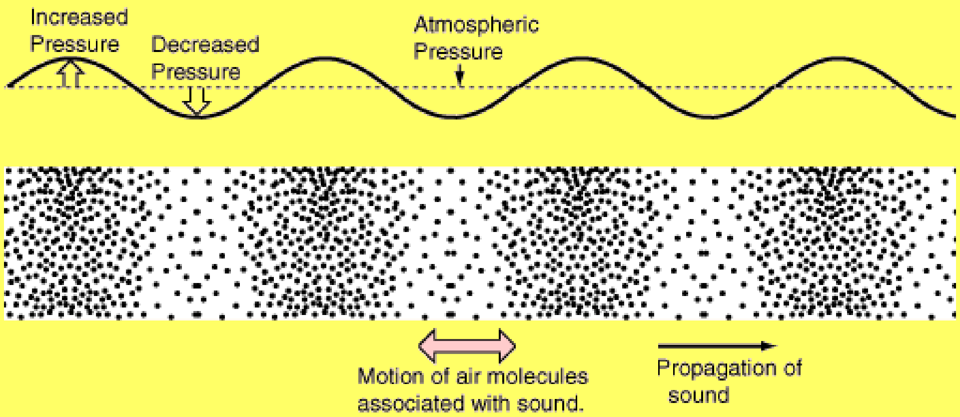
causes the adjacent air molecules to be driven back and forth,

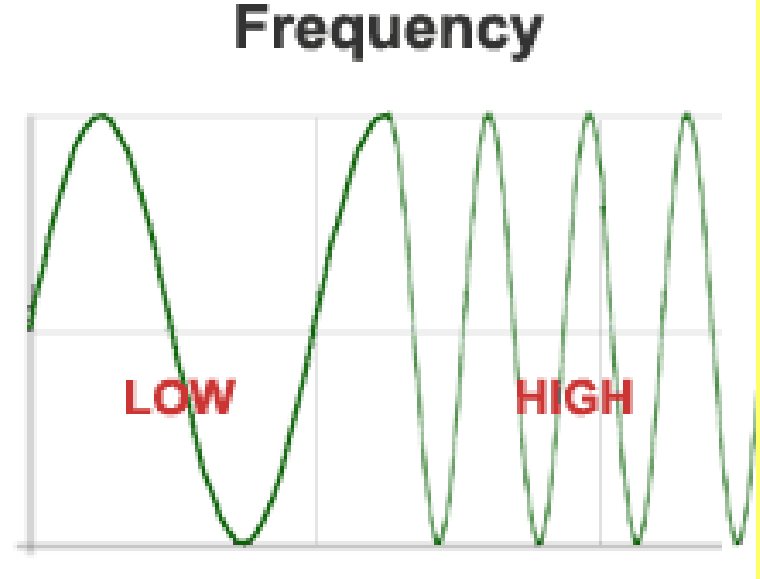
in turn driving the next molecules and so on…

producing a series of compressions (slight increase in pressure) and rarefactions (slight decreases in pressure)

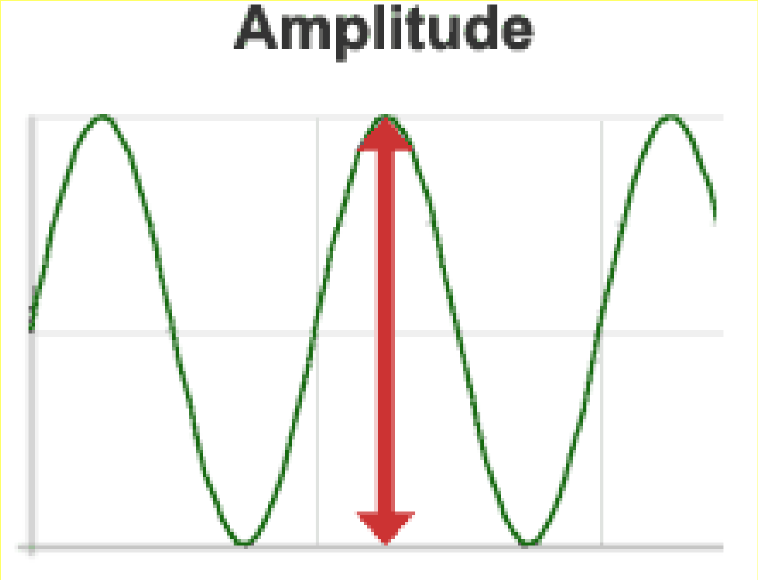
Tuning fork

Sine wave

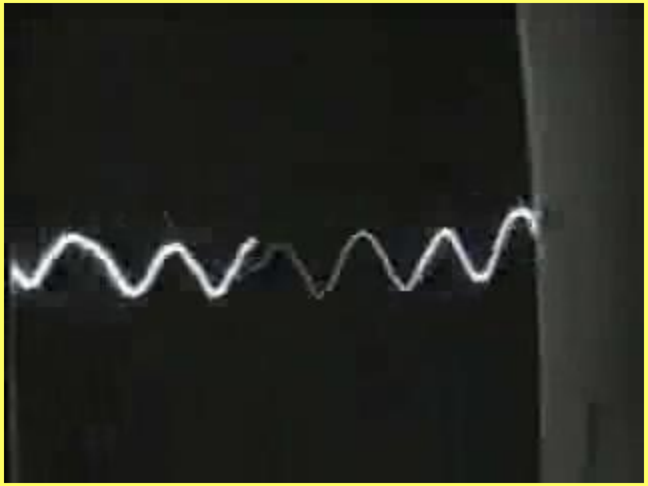

Increased Pressure
Decreased Pressure
Atmospheric Pressure
Motion of air molecules associated with sound.
Propagation of sound



Frequency: cycles per second = hertz (Hz)

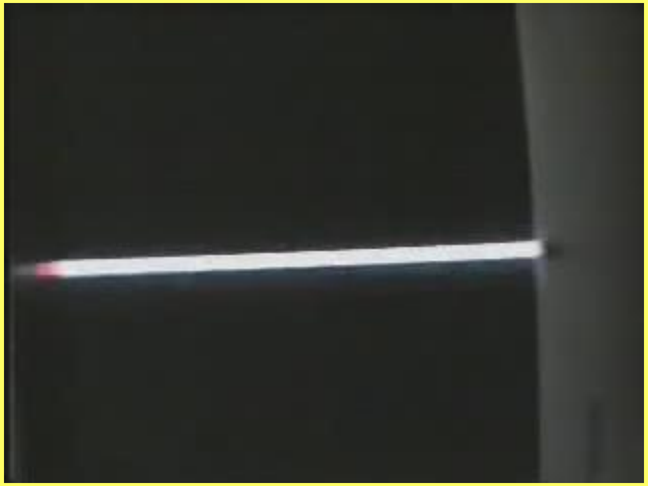

Amplitude

Simple waveform



Complex waveform

Whitney



Why we hear

- Things
  - things to eat
  - things that might eat us
  - inanimate dangers

- People
  - non speech communication
  - speech communication
  - SPEECH

What is speech

- vocalized form of human communication.
- based upon a structured combination of words and names drawn from..
- a very large vocabulary where..
- each spoken word is created out of the combination of a limited set of vowel and consonant speech sound units..
- the vocabularies, the syntax which structures them, and their set of speech sound units differ producing many mutually unintelligible human languages

English

250,000 words (OED) but 10,000? in everyday use

4000 syllables (usually 1 vowel + 1 or 2 consonants)

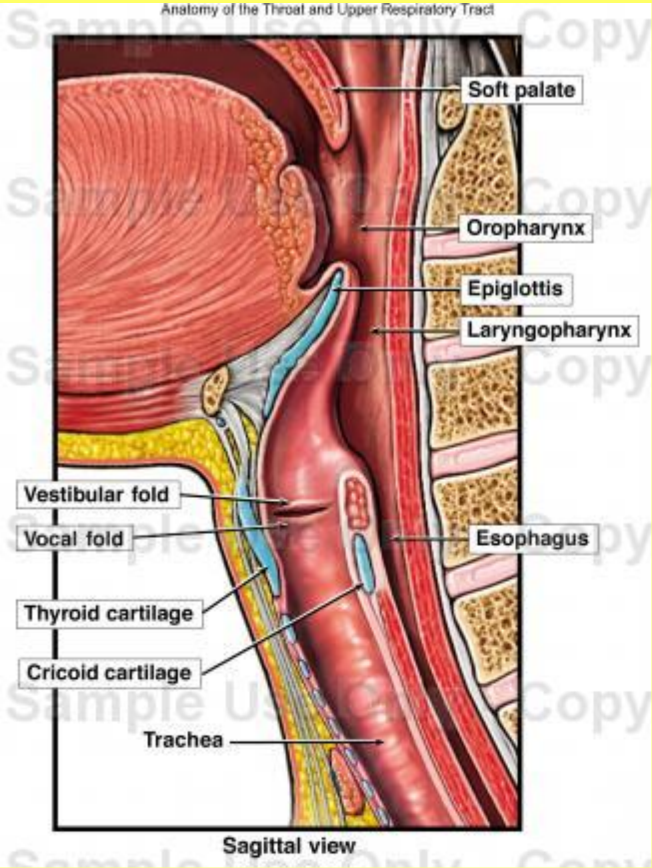
eg “water” has 2 syllables: *wa* and *ter*

40 – 50 phonemes, the smallest unit of speech sound that differentiates one word from another.

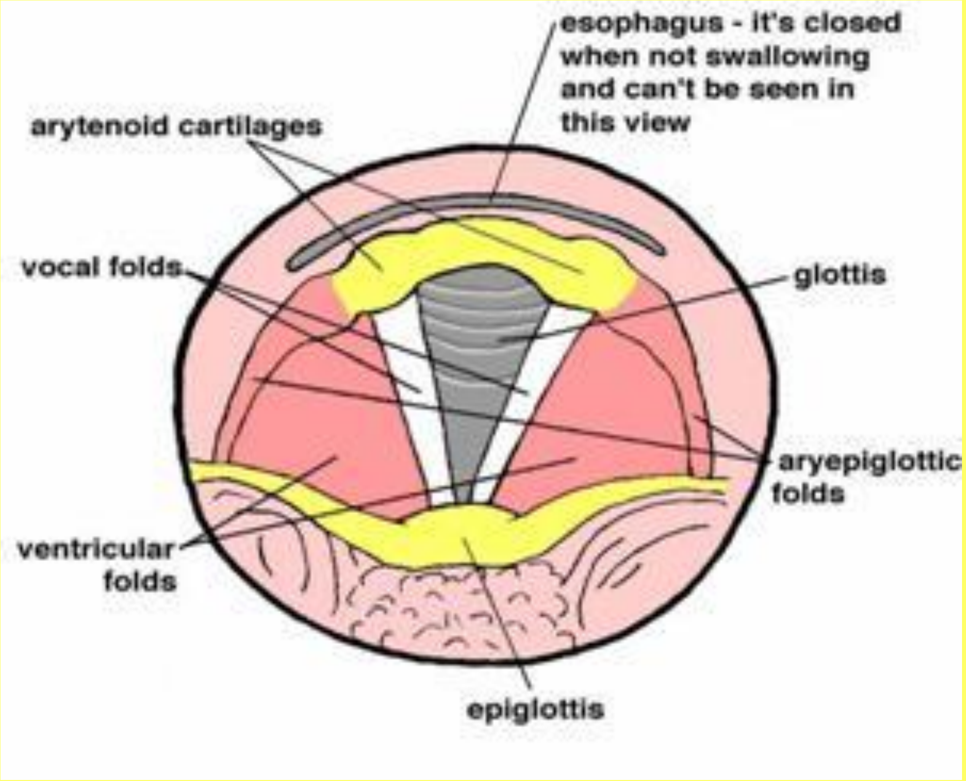
Pat and bat each have one syllable but 3 phonemes

Speech production

- Bellows: the lungs
- Vibrator: the larynx
- Modifiers:
  - tongue,
  - palate
  - jaws
  - lips



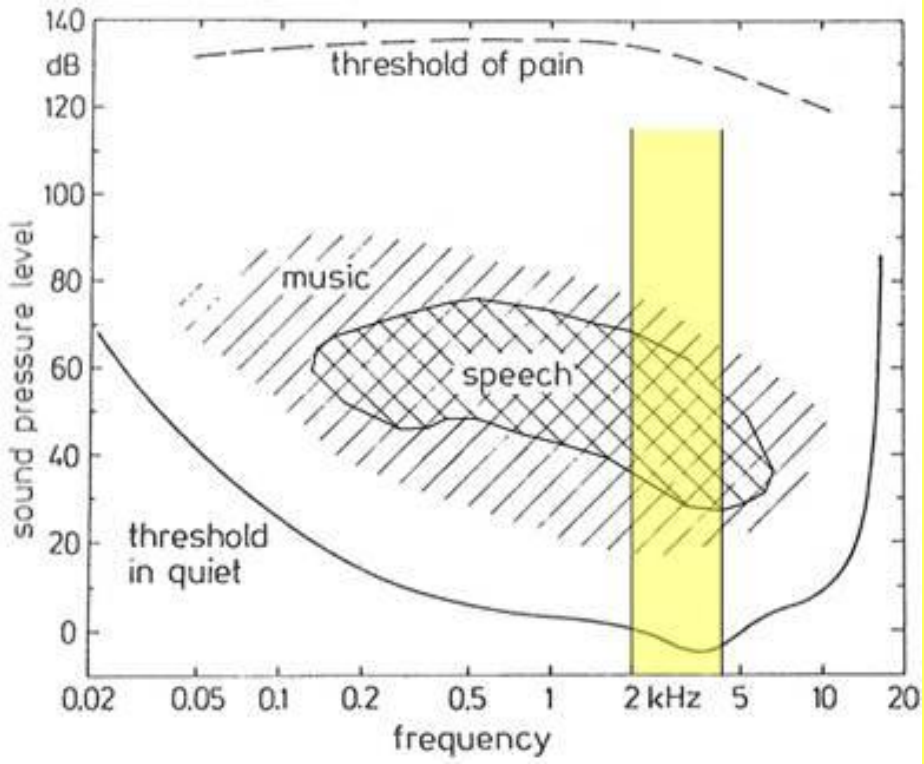
Speech: the larynx



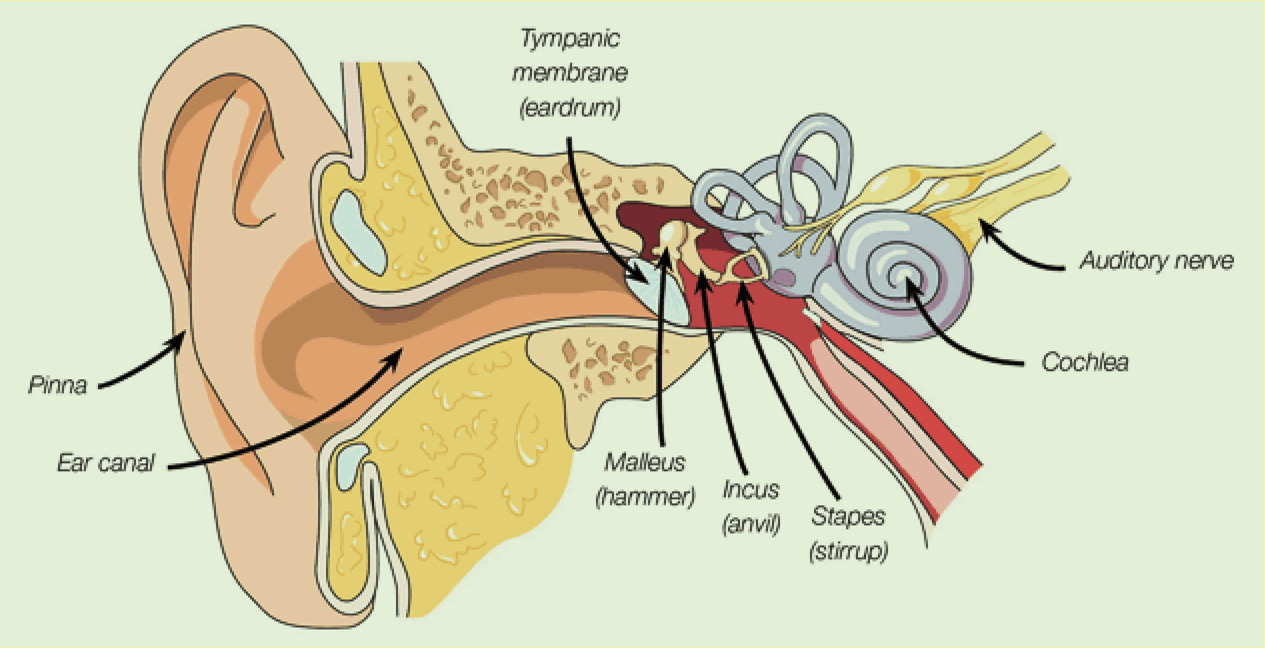
Speech: the real larynx

Speech: the modifiers

The sounds of speech

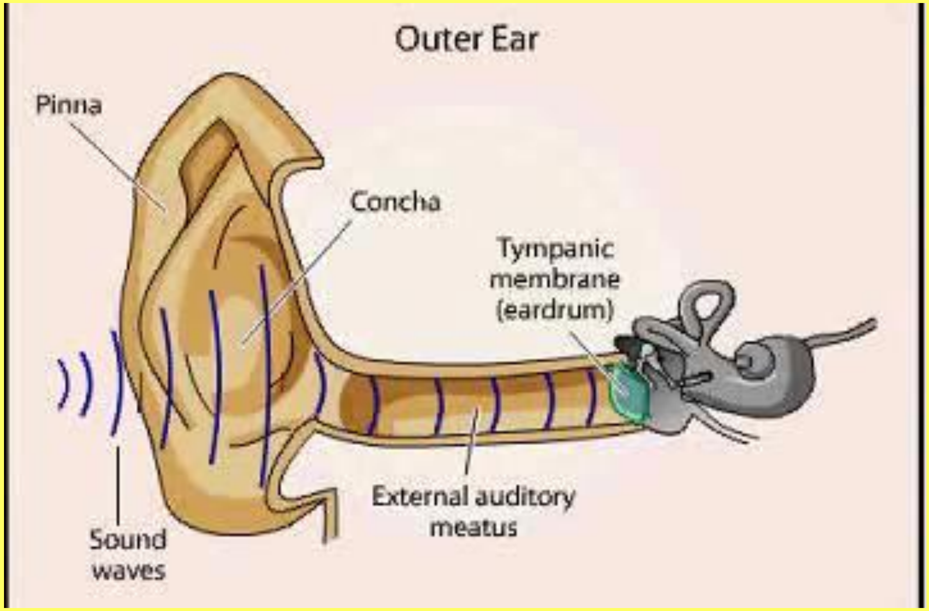


How we hear

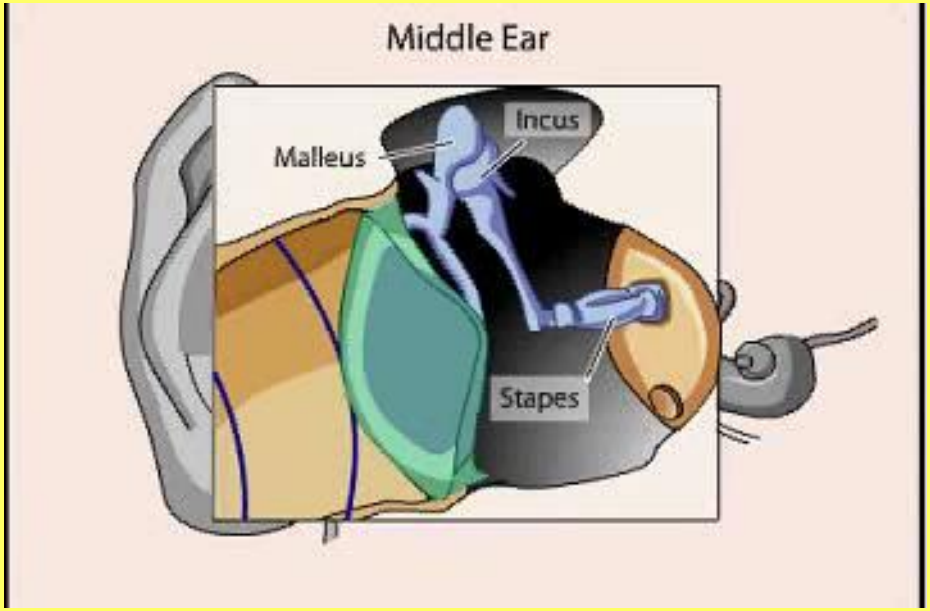


Tympanic
membrane
(eardrum)
Auditory nerve
Cochlea
Pinna
Ear canal
Malleus
(hammer)
Incus
(anvil)
Stapes
(stirrup)

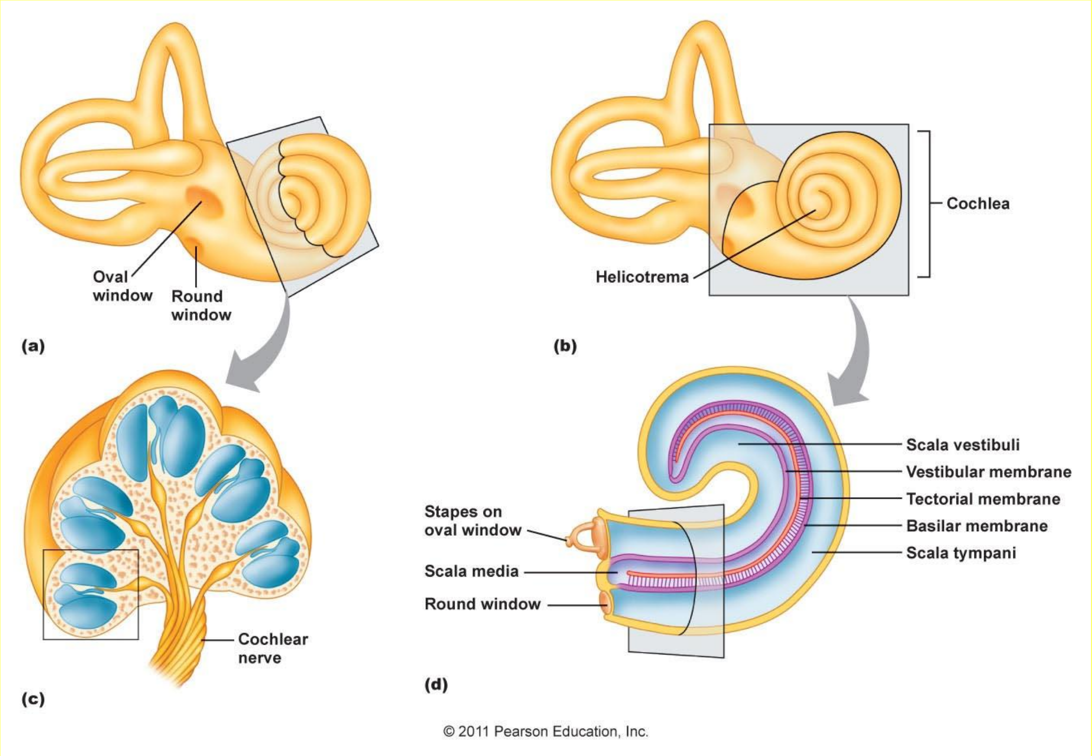
Outer ear


Outer Ear
Pinna
Concha
Tympanic
membrane
(eardrum)
External auditory
meatus
Sound
waves

Middle ear

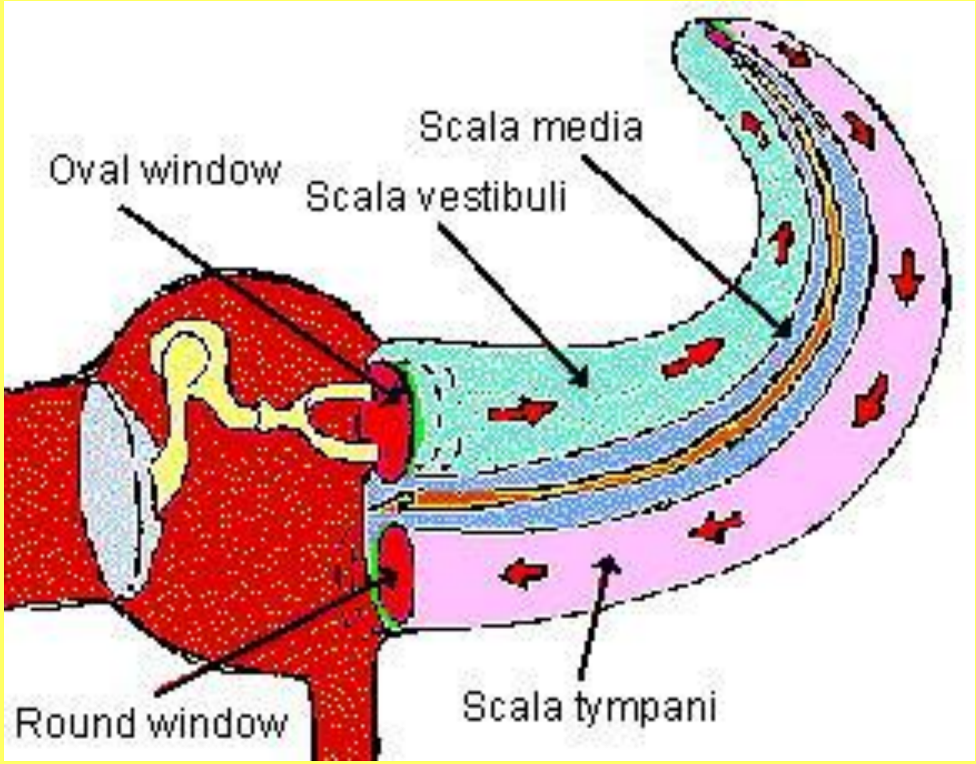


Inner ear

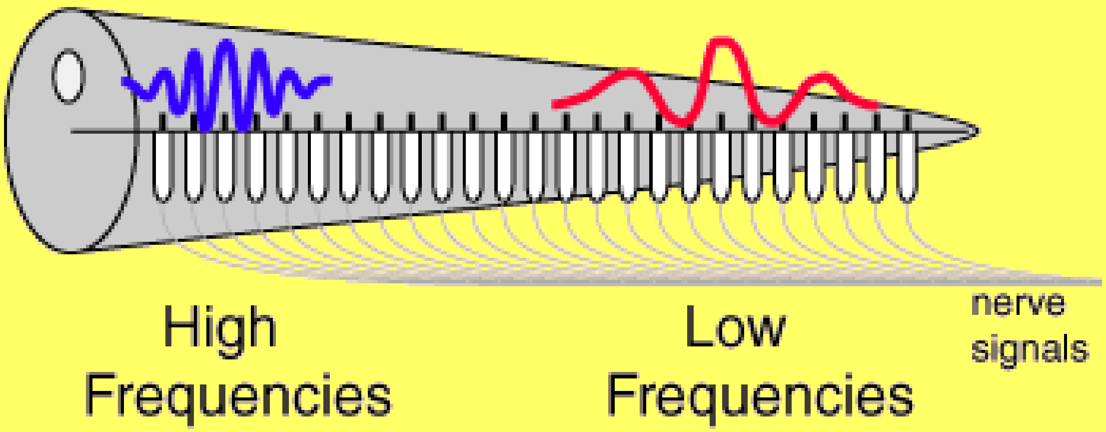


Oval window
Round window
(a)
Cochlea
Helicotrema
(b)
Cochlear nerve
(c)
Scala vestibuli
Vestibular membrane
Tectorial membrane
Basilar membrane
Scala tympani
Stapes on oval window
Scala media
Round window
(d)
© 2011 Pearson Education, Inc.

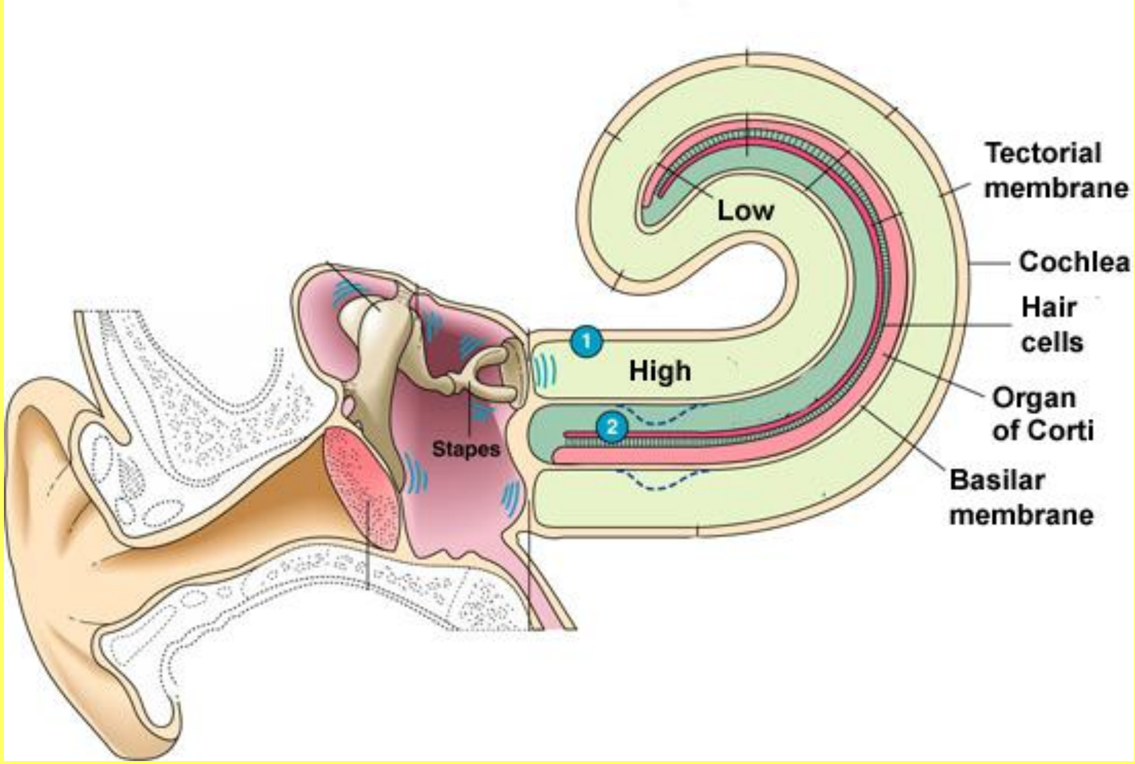
The cochlea


Scala media
Oval window
Scala vestibuli
Round window
Scala tympani

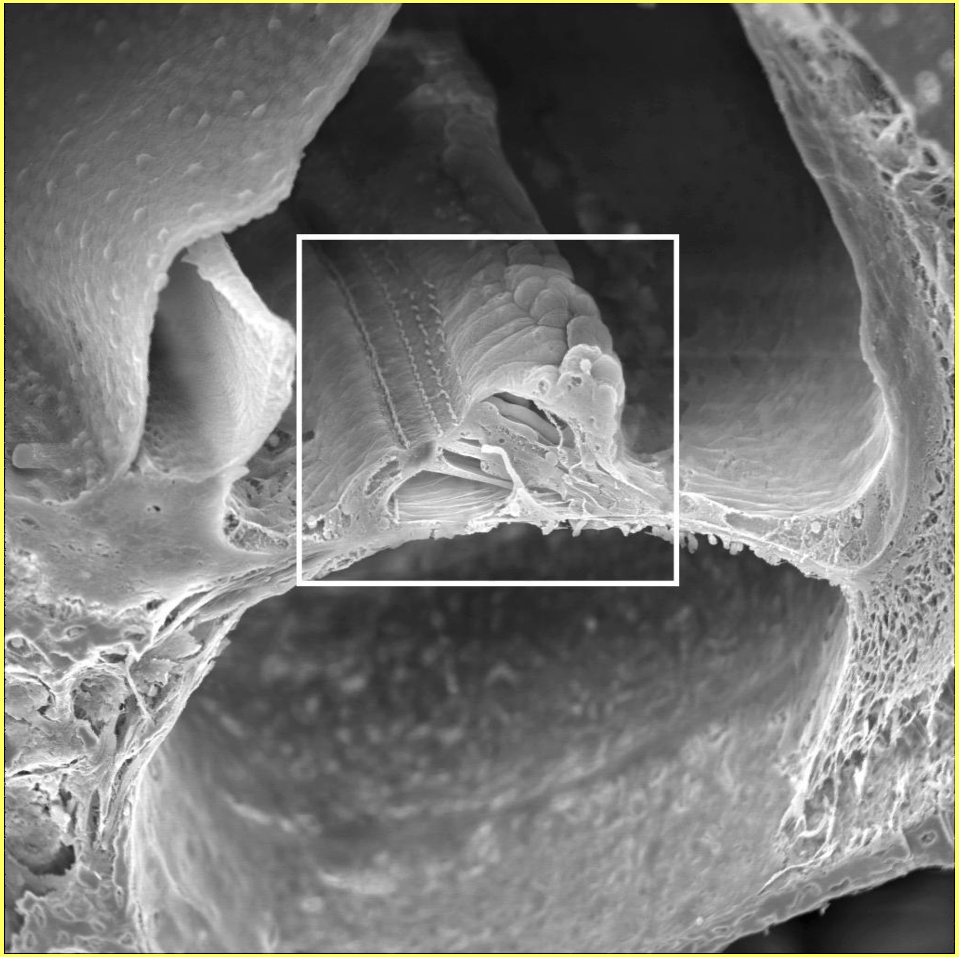
Cochlea is physically tuned



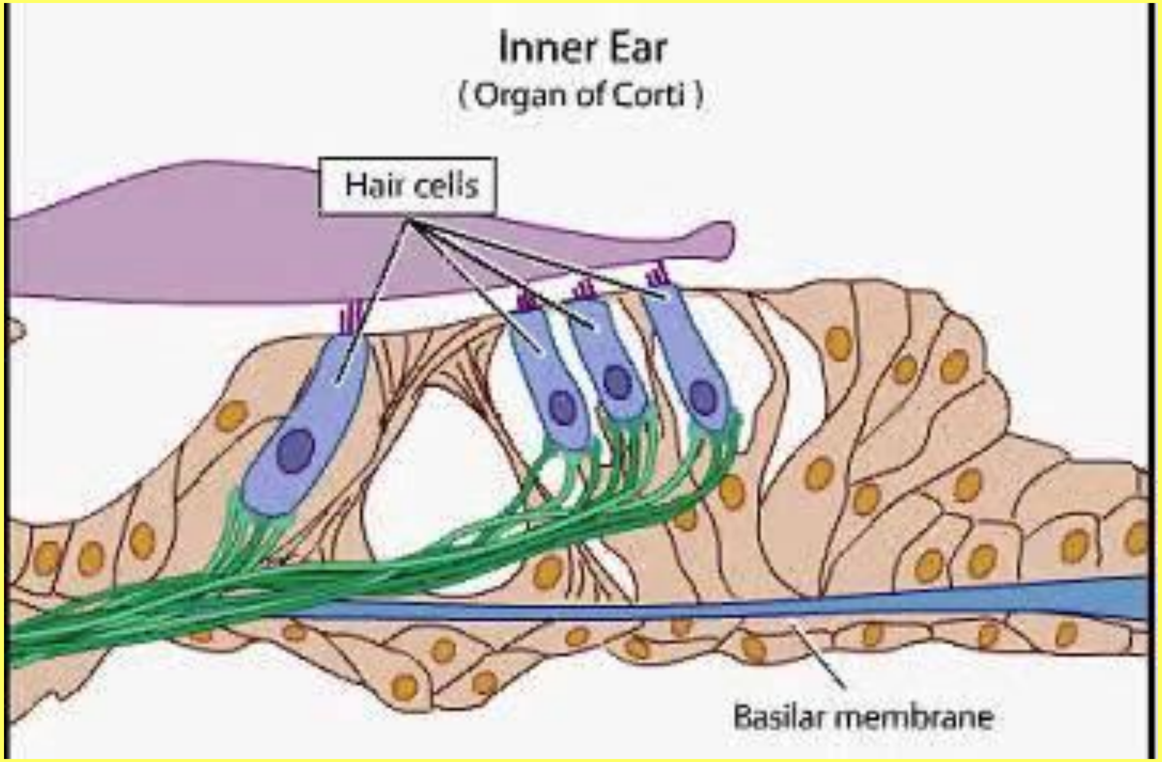
from high to low



The inner inner ear

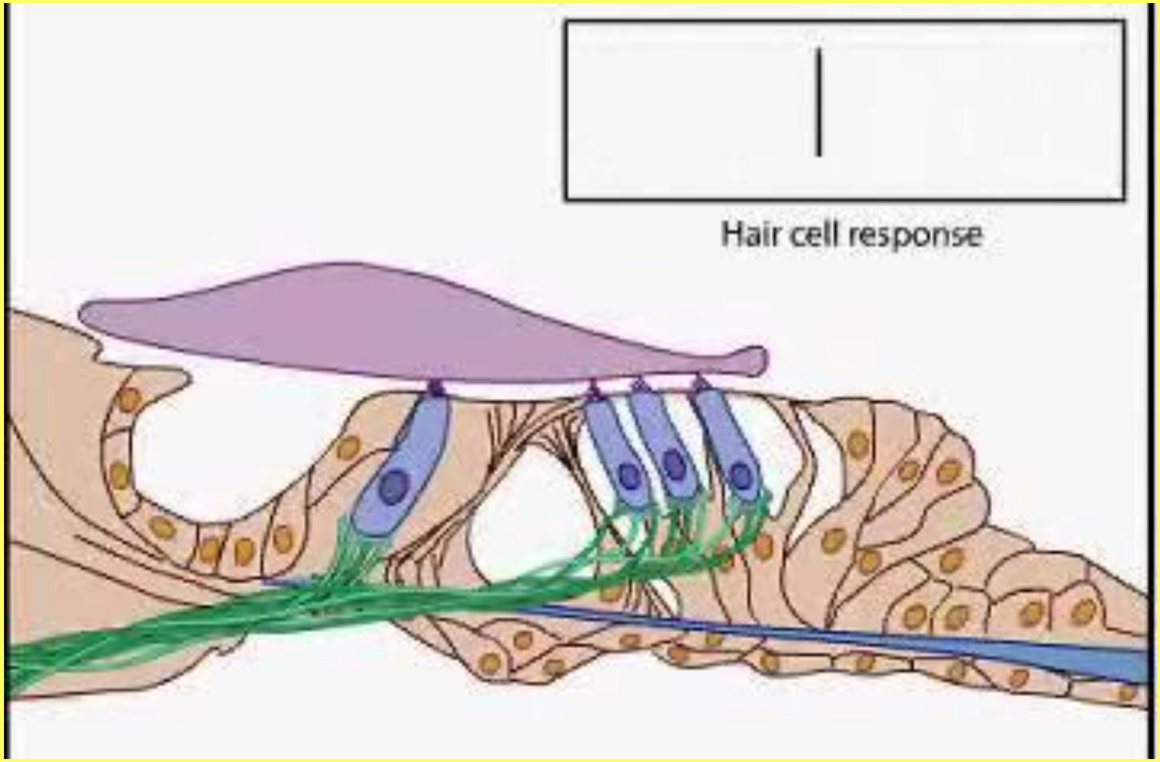


The hair cells

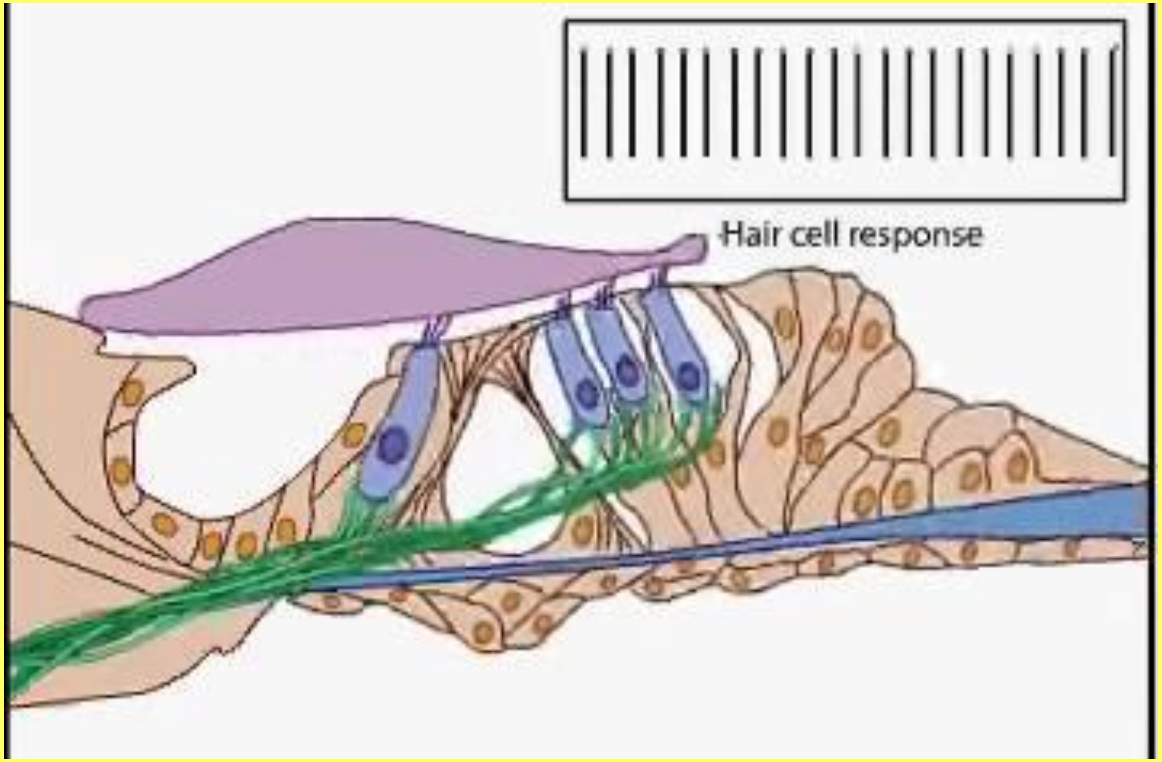


The hair cells

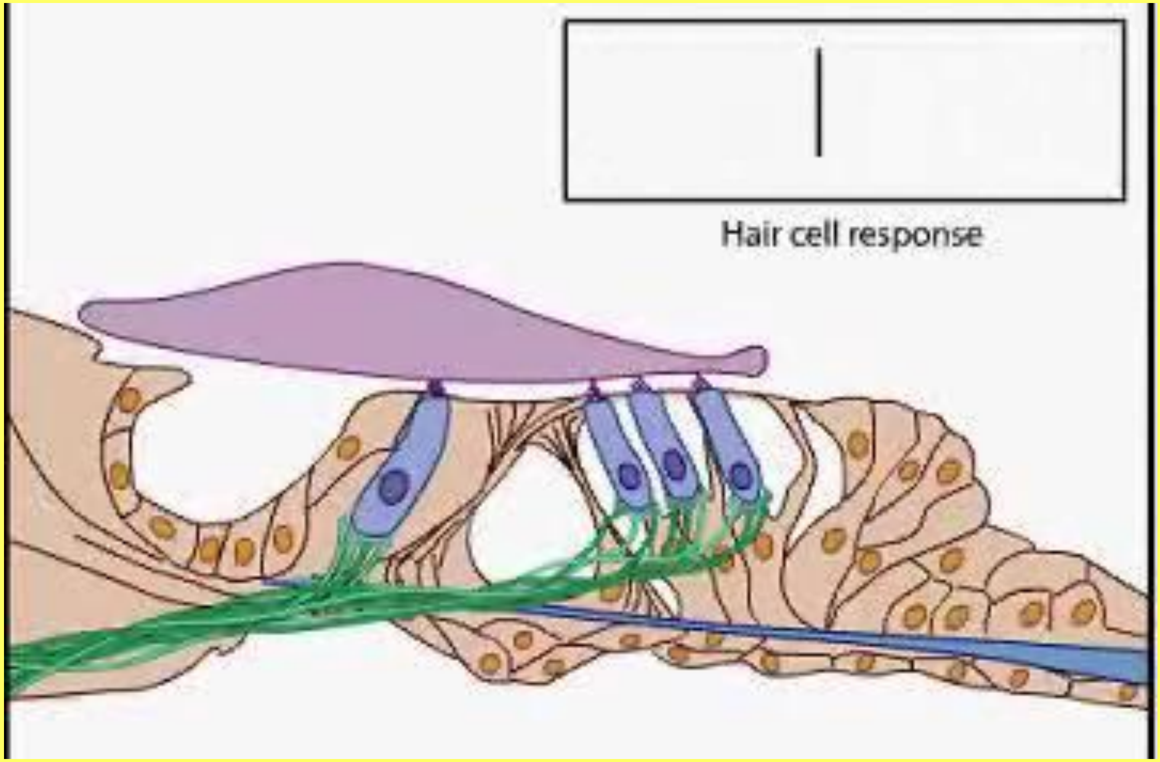
Hair cell movement



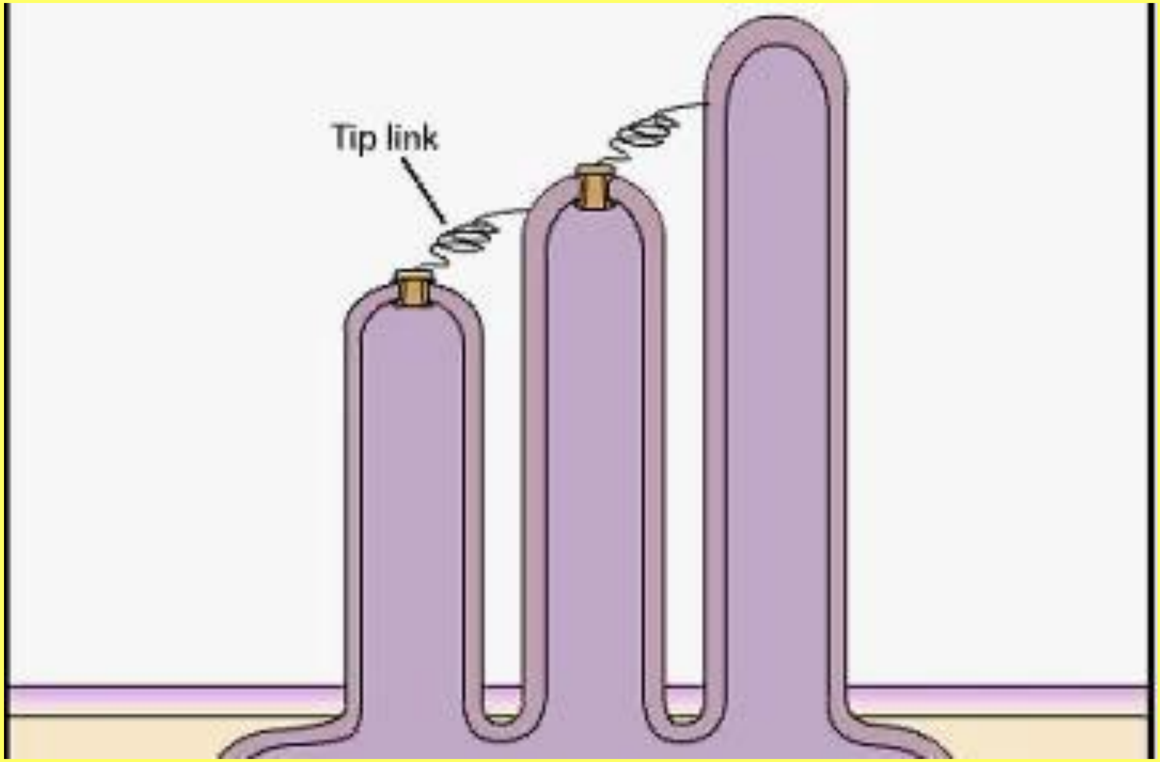
up

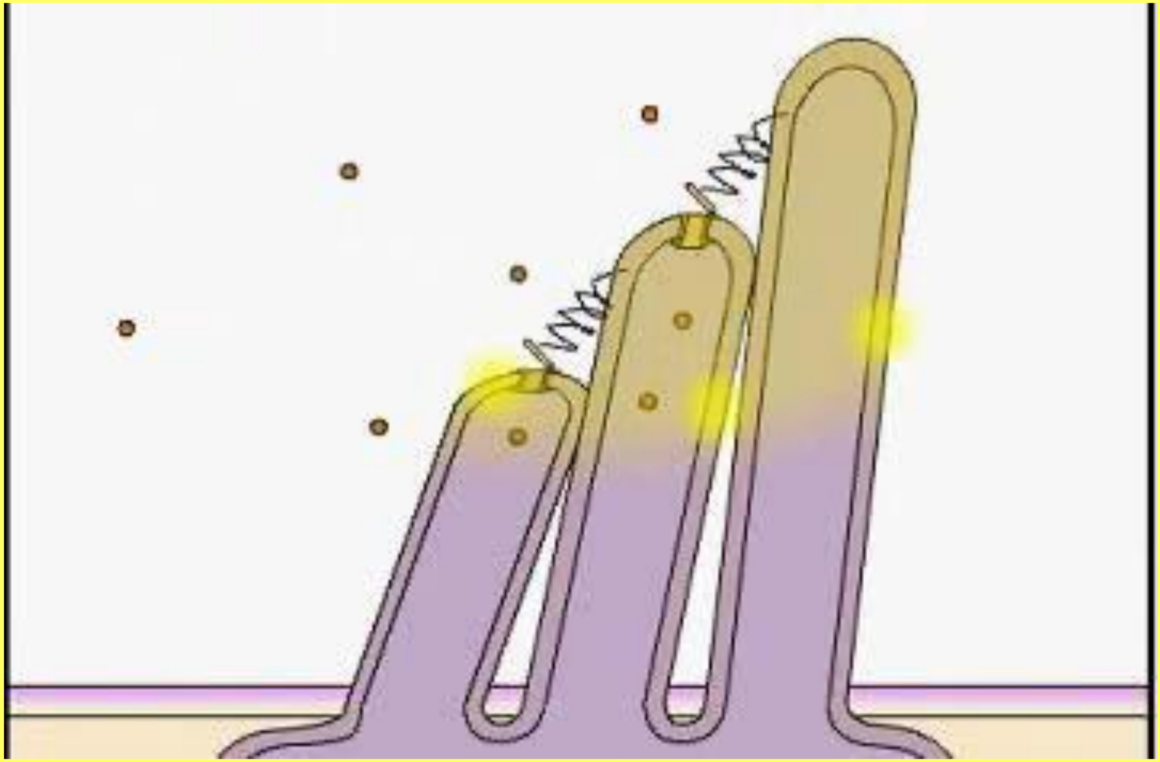


down



The “hairs”





How good is human hearing?

	Frequency range		
• Elephant	16 Hz	-	12,000 Hz
• Human	20 Hz	-	20,000 Hz
• Cow	23 Hz	-	35,000 Hz
• Cat	45 Hz	-	64,000 Hz
• Mouse	1000 Hz	-	91,000 Hz

How good is human hearing?

Sound localisation

- requires binaural hearing
- makes use of interaural time and intensity differences
- Can detect 0.5dB interaural loudness difference
- 10 millionth of interaural time difference
- Which equates to being able a left right difference of 1 degree

How good is human hearing?

Intensity

- Softest sound audible causes movement of inner ear membranes of 0.2nm – diam of 2 hydrogen atoms
- A whisper is 1000x more powerful than a just audible sound
- Conversational speech is 1,000,000x more powerful
- A loud shout 1,000,000,000x more powerful
- The full dynamic range from just detectable to just tolerated is 1:10,000,000,000,000 (10^{13})

decibels

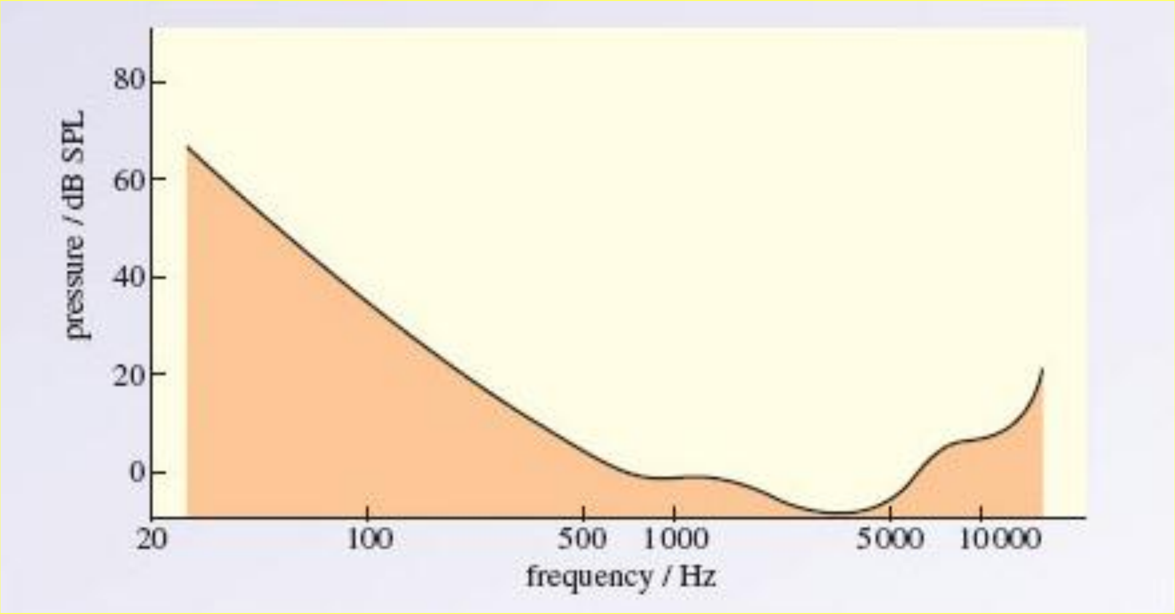
- Measuring very large range
- Use the logarithm of the ratio of 2 values

  1 bel = $\log^{10} I_1 / I_2$

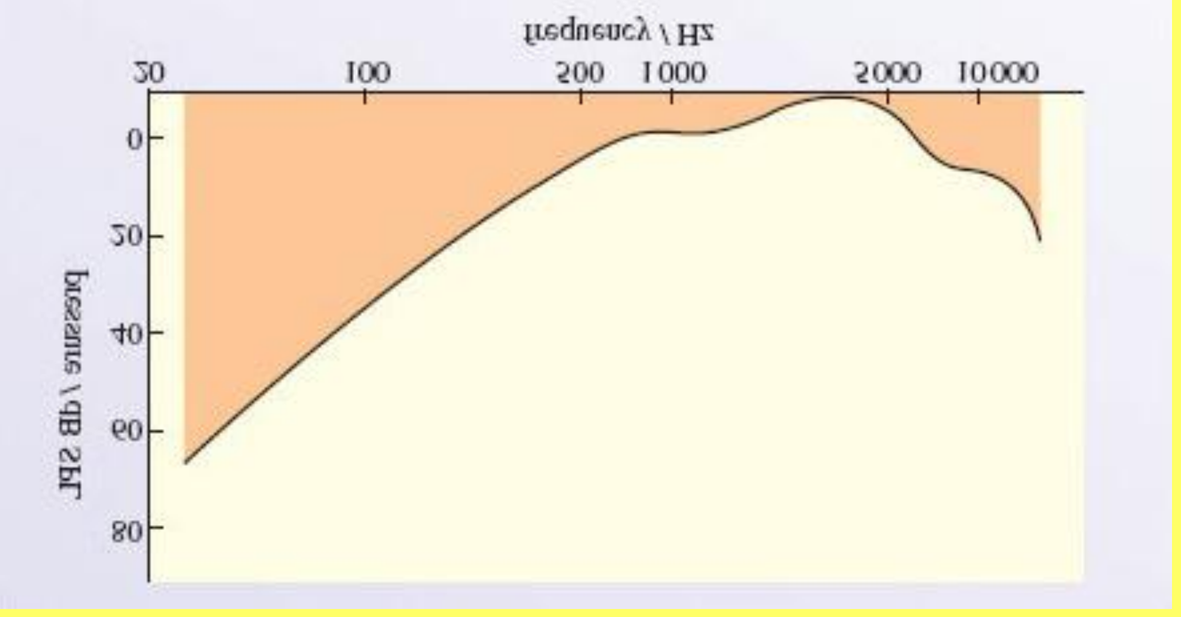
  1 decibel (dB) = $10 \log^{10} I_1 / I_2$

- Dynamic range of 1 : 10,000,000,000,000 (10^{13})
- now becomes 1 : 130 dB

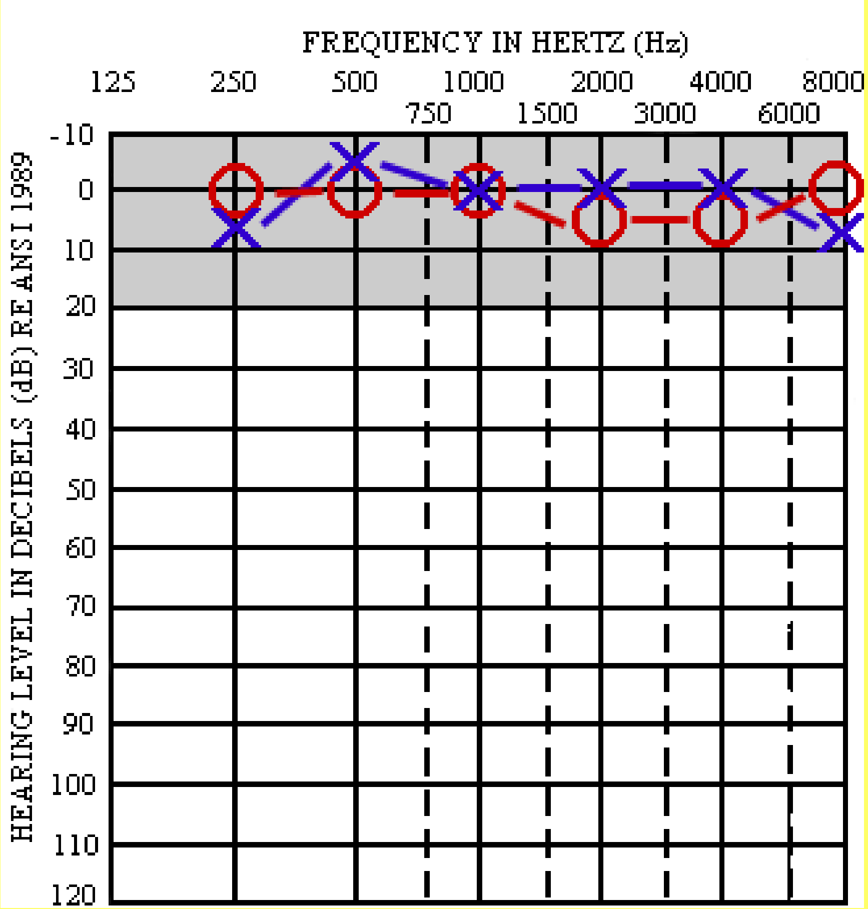
Threshold of hearing



Testing hearing



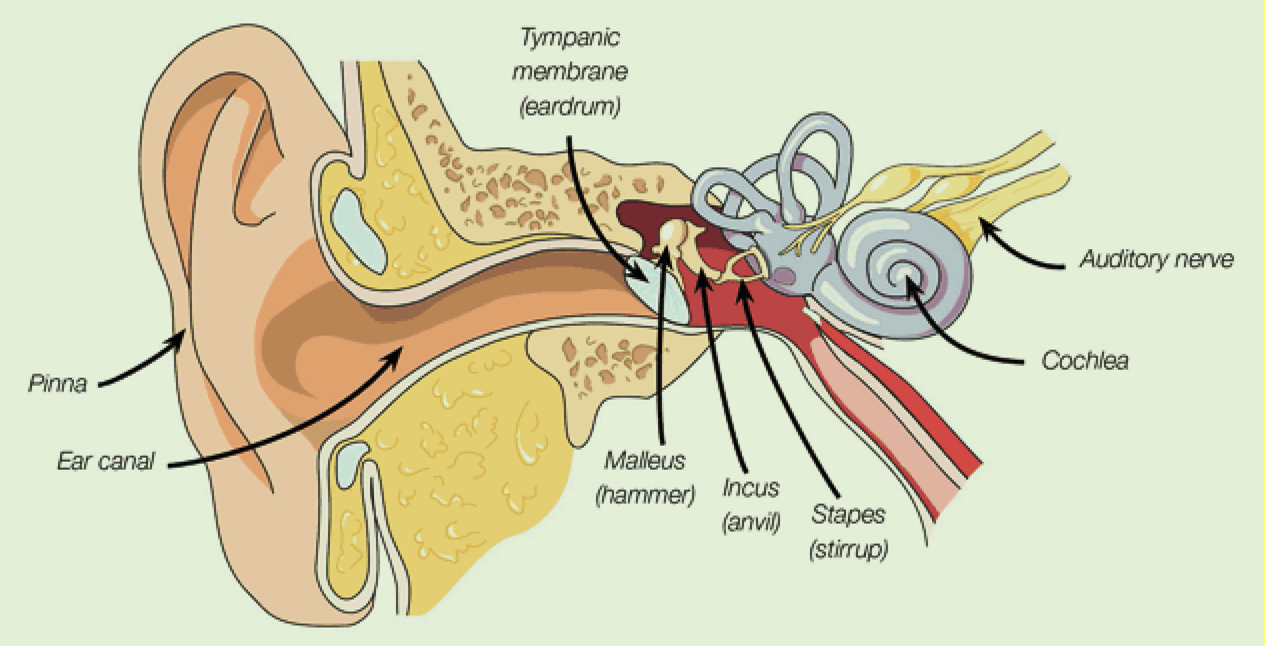
Pure Tone Audiometry



Speech audiometry

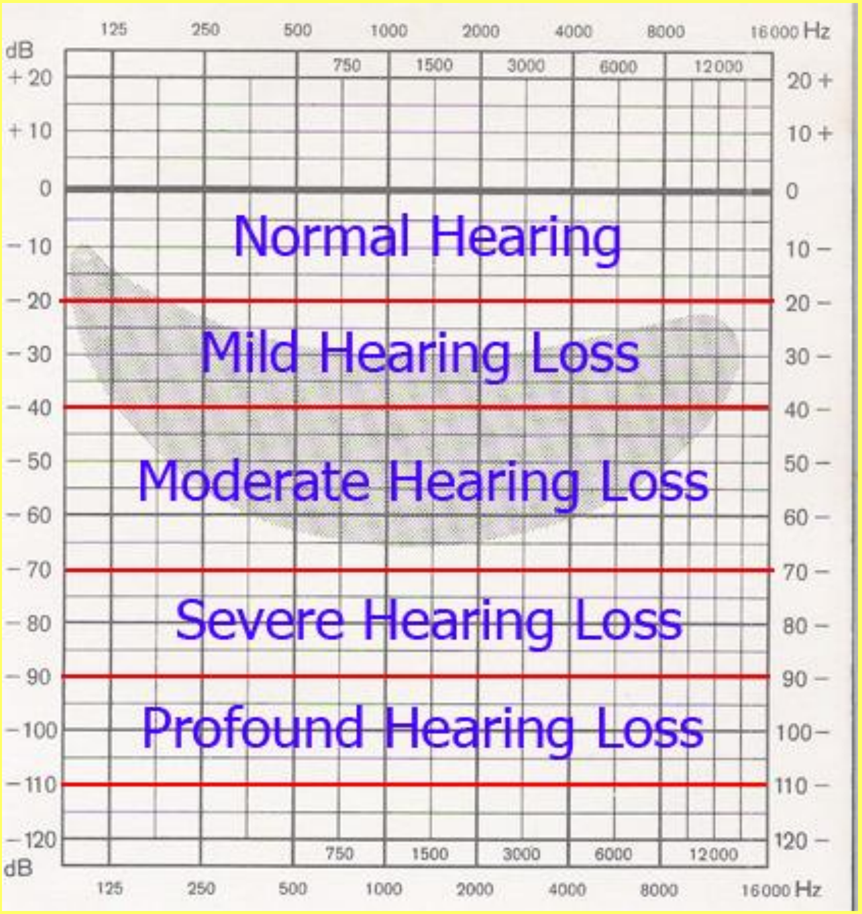
- Voice tests
- Speech audiometry (phonemes)

Deafness



Tympanic
membrane
(eardrum)
Auditory nerve
Cochlea
Pinna
Ear canal
Malleus
(hammer)
Incus
(anvil)
Stapes
(stirrup)

Severity of deafness



Prevalence of deafness

Age	mild	mod	severe	total
0-14	0.1	0.1	0.06	0.25
15-50	2.3	0.4	0.1	2.8
51-60	13.9	2.6	0.5	17.0
61-70	39.7	6.7	1.5	47.8
71+	54.7	6.4	2.5	63.6
total	10.8	1.6	**0.4**	12.9

% of population (22m)

Listen Hear 2006 (Australia)

Cost of deafness

- A$11.75 billion, 1.4% GDP
- A$6.7 billion productivity loss
- Informal carers
- Direct health costs

- A$11.3 billion Quality of life impacts

What to do about deafness

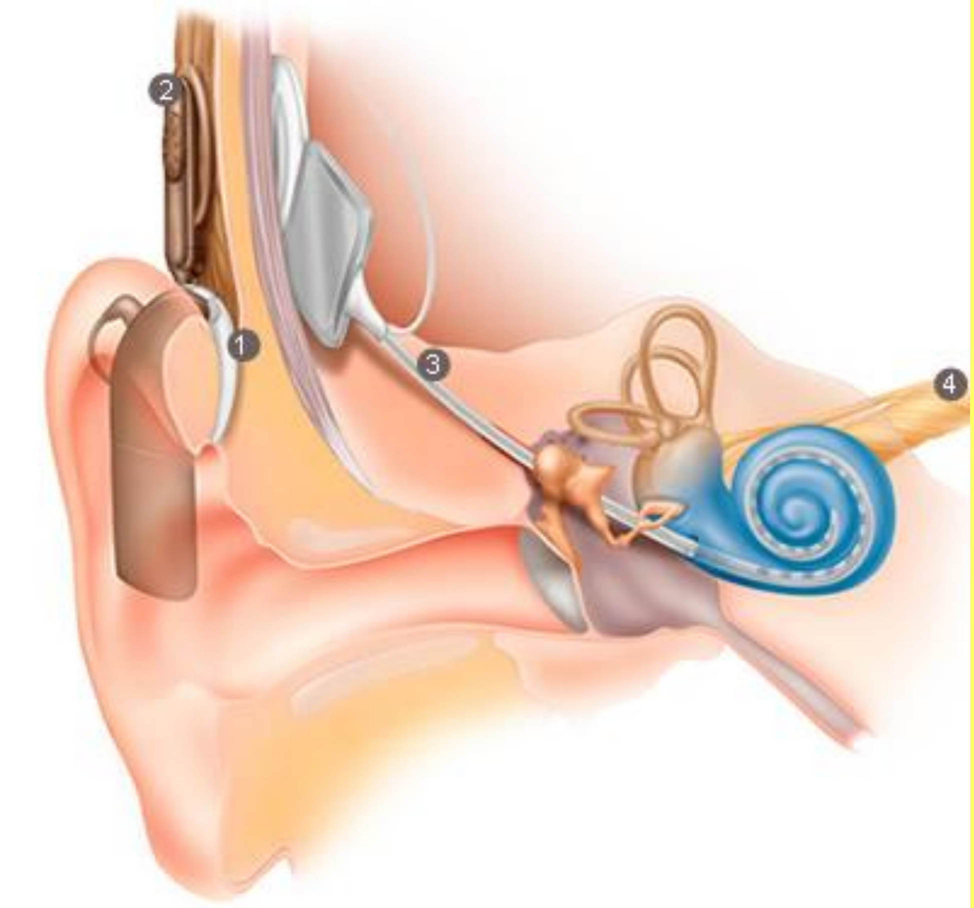
- Pretend everything normal
- Hearing tactics
- Hearing aids

- Cochlear implant
- Brainstem implants
- Hair cell regeneration

Cochlear implant

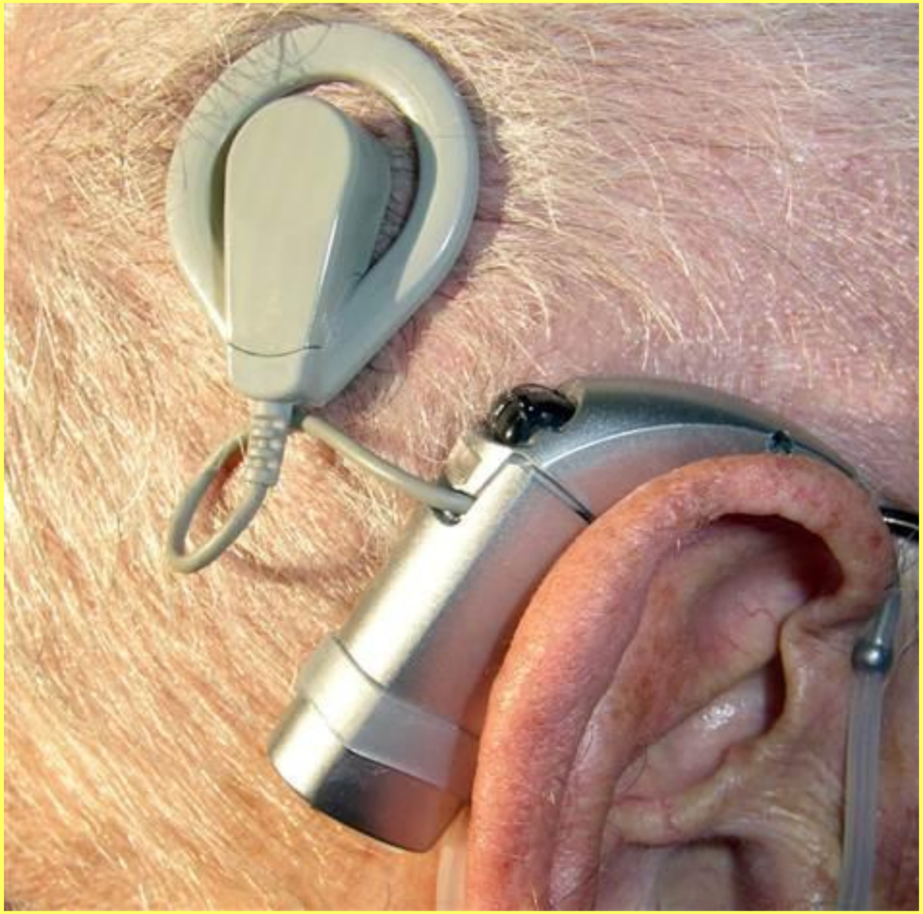
- electronic device that stimulates the cells of the auditory nerve to produce a sense of sound in a person.
- internal implanted component
- externally worn component with a microphone and speech processor which transmits information and power to the internal component wirelessly.

Cochlear implant



CI internal

CI external



CI wearer

Who is suitable for CI

NICE 2009

- Unilateral CI for people with severe to profound deafness who do not receive adequate benefit from acoustic hearing aids
- 90 dB or worse at 2000Hz and 4000Hz
- less than 50% aided speech discrimination

Who is suitable for CI

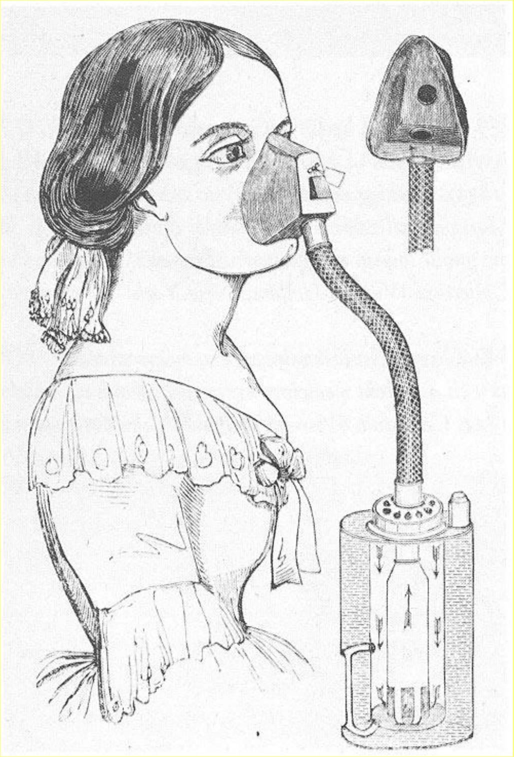
NICE 2009

Simultaneous CI for following groups of people with severe to profound deafness who do not receive adequate benefit from acoustic hearing aids

- Children
- Adults who are blind or who have other disabilities that increases their reliance on auditory stimuli as a primary sensory mechanism for spatial awareness

How do we get it in

- General anaesthetic



How do we get it in

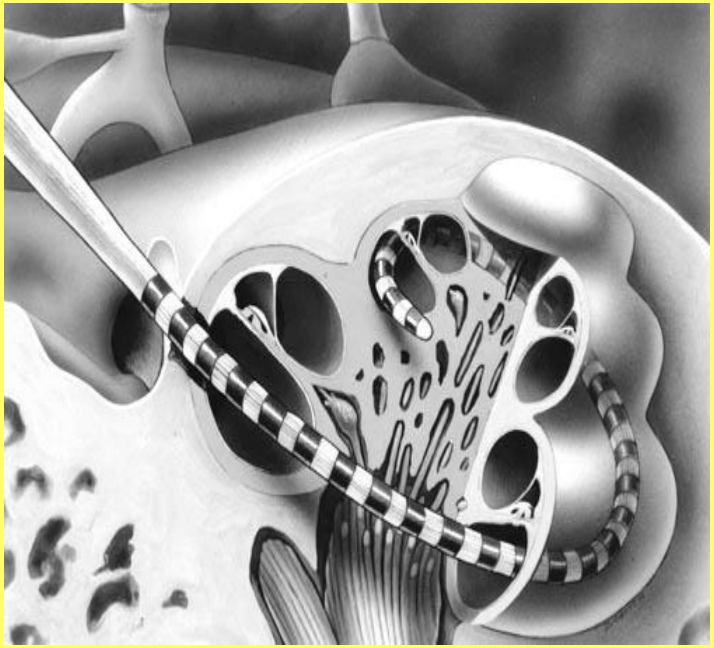
- General anaesthetic
- Surgical instruments

How do we get it in

- General anaesthetic
- Surgical instruments
- Operating microscope

How do we get it in

- General anaesthetic
- Surgical instruments
- Operating microscope
- Drill a hole through the head and into the cochlea



How good are cochlear implants?

- Still unpredictable
  - Age of onset of deafness (pre/post lingual)
  - Length of deafness
  - Use of oral communication or signing
  - Adequacy of previous hearing aids
  - cause of deafness

How good are cochlear implants?

- Nearly all have some perception of sound
- Presence / absence of sound helps lip reading
- Most get some discrimination of pitch and loudness
- Some able to understand speech without lip reading
- Some able to use telephone

How good are cochlear implants

- No implantee has normal thresholds (best are 25-30 dB)
- All dependent on externally worn device

What does an implant sound like?

Simulated CI sound perception using

1, 4, 16 and 22 channels

and music

Simulated music heard through

4 and 32 channel implant

Many CIs function at the 8 – 16 level

rhapsody

Risks of cochlear implantation

- General anaesthesia
- Risks of the surgery
  - Bleeding
  - Infection
  - Meningitis
  - Vertigo
  - Facial nerve injury

Risks of cochlear implantation

- General anaesthesia
- Risks of the surgery
- Risks associated with the implant
  - Mechanical or electrical failure
  - Rejection
  - Infection
  - Removal and or replacement

Reversibility of myopia

- Nothing
- Monocle
- Spectacles
- Contact lenses
- Laser refractive surgery

Reversibility of a lost hand

Is deafness reversible

- Impairment
- Disability
- Handicap

Does CI reverse deafness?

Yes

Some people can hear and understand speech using CI alone

Does CI reverse deafness?

No

- Not suitable for every deaf person
- Dependant on externally worn device
- Not as good as an ear for speech
- Not nearly as good as an ear for music
- Currently no spatial awareness
- Functionally at best a very scratched monocle?

Merry Christmas

and

Happy new year